

The Frontend phase transforms your architecture and requirements into detailed user interface and user experience specifications. This section focuses on creating comprehensive frontend specifications that deliver exceptional user experiences while maintaining technical excellence.

What You'll Learn

In this frontend phase, you'll:

- **Create design systems** with comprehensive UI/UX specifications and design tokens
- **Design component architecture** with React patterns and state management
- **Implement performance optimization** with PWA features and bundle optimization
- **Ensure testing and accessibility** with comprehensive validation and WCAG compliance

Frontend Exercises

1. Design System and UI/UX

Create comprehensive design system specifications including visual design, interaction patterns, responsive layouts, and accessibility requirements.

2. Component Architecture

Design detailed React component architecture with TypeScript, state management patterns, and reusable component libraries.

3. Performance Optimization

Implement performance optimization strategies including bundle optimization, caching, PWA features, and Core Web Vitals compliance.

4. Testing and Accessibility

Create comprehensive testing strategies and accessibility compliance ensuring WCAG 2.1 AA standards and cross-browser compatibility.

Key Benefits

By completing the Frontend phase, you'll have:

- **Comprehensive Design System:** Consistent visual design and interaction patterns
- **Scalable Component Architecture:** Maintainable React components with TypeScript
- **Optimized Performance:** Fast-loading application with excellent user experience
- **Accessible and Tested:** WCAG compliant application with comprehensive test coverage

Kiro Features Integration

Throughout this phase, you'll leverage:

- **React Design Systems MCP Server:** For AWS Cloudscape best practices and component guidance
- **Web Search MCP Server:** For current frontend best practices and design trends
- **NPM Registry MCP Server:** For package research and dependency management
- **Agent Hooks:** For code quality validation and documentation updates
- **Agent Steering:** For frontend standards and React best practices

Prerequisites

Before starting the Frontend exercises:

- Foundation and Infrastructure phases completed
- Understanding of React and TypeScript
- Familiarity with modern frontend development tools
- Knowledge of web accessibility principles

Let's begin with creating a comprehensive design system that will guide all visual and interaction design decisions throughout your chatbot application.